Hemoglobin Degradation& Jaundice

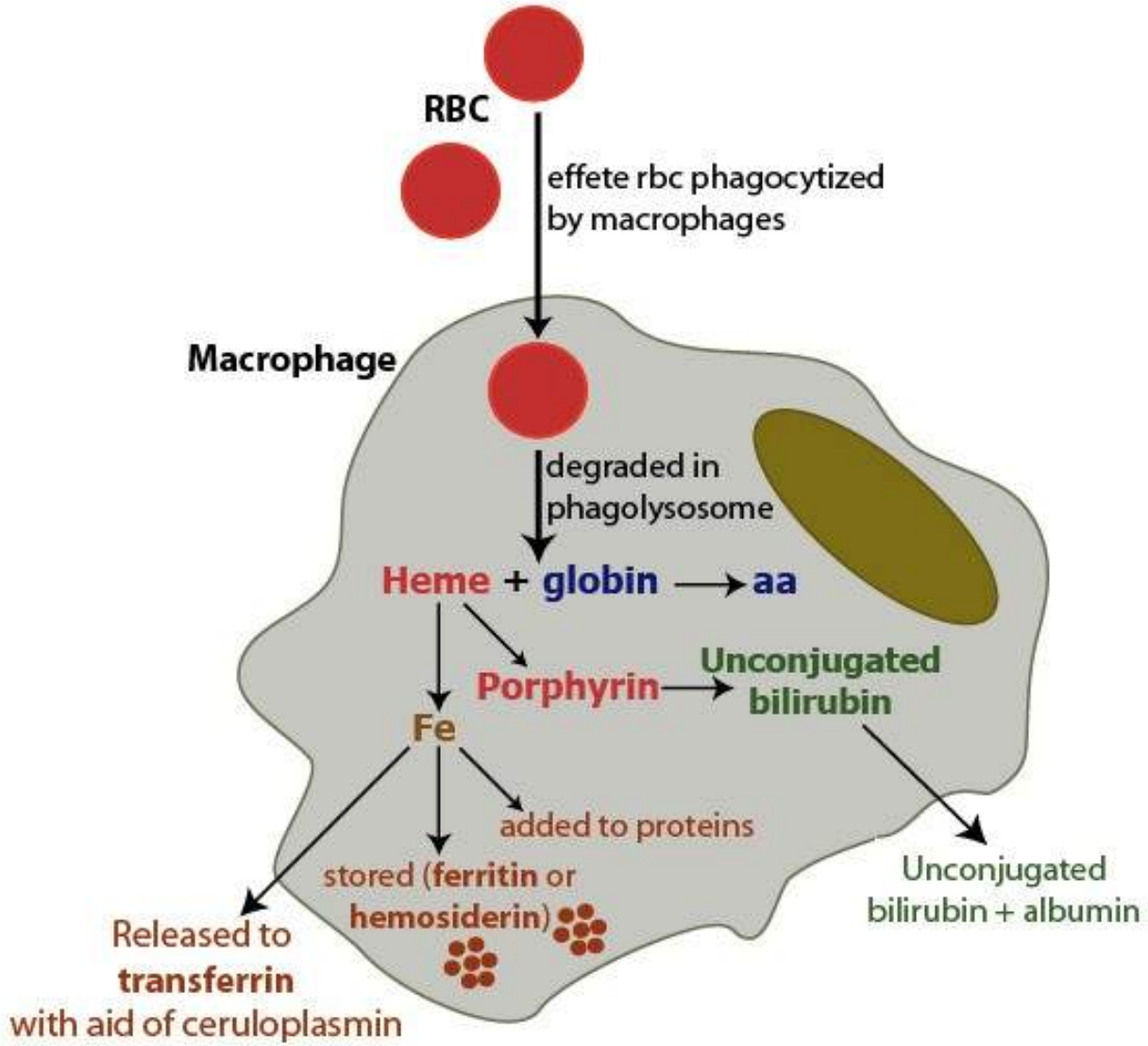
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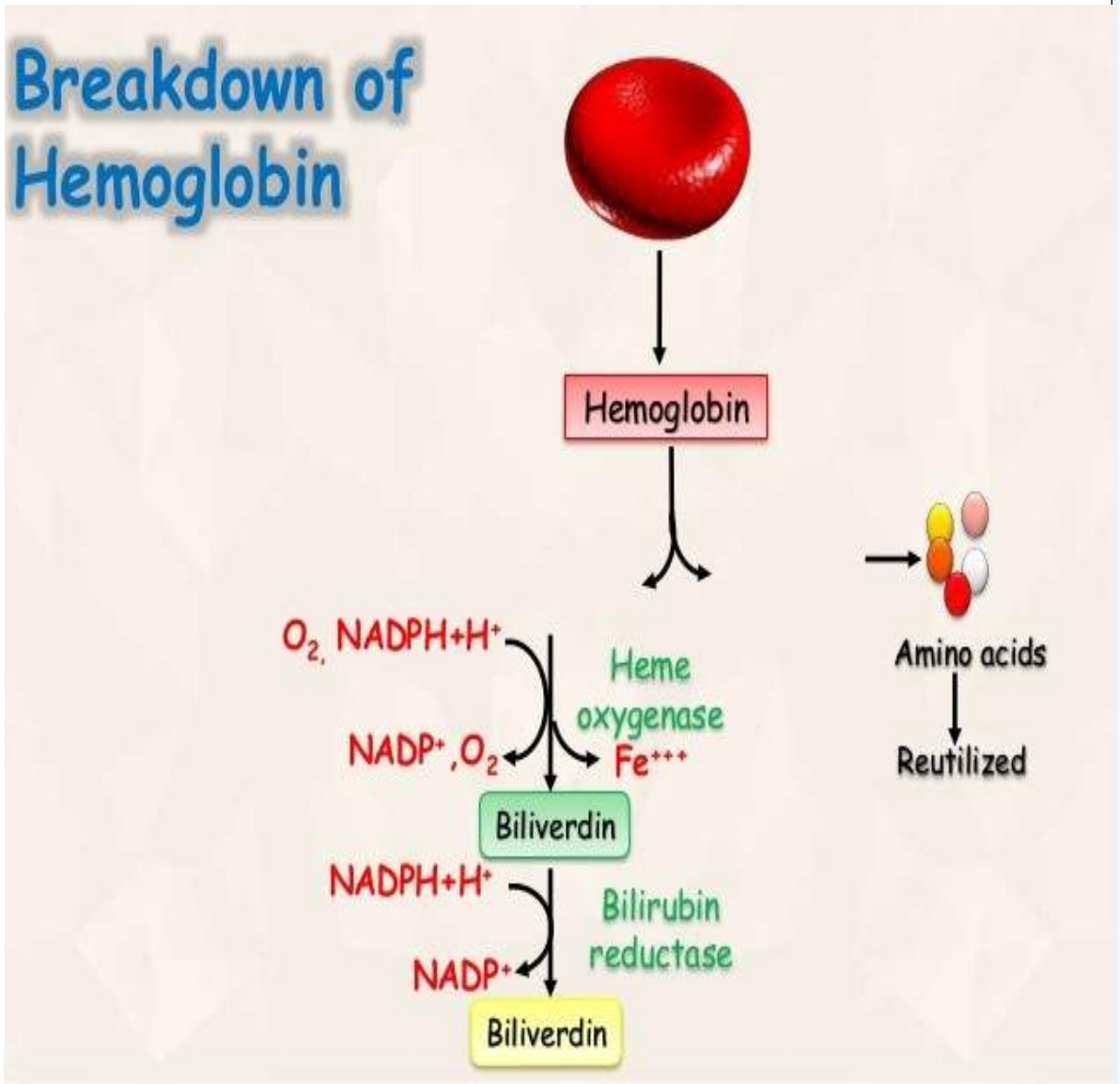
Objectives:

- Define heme degradation
- Describe the process of Hb degradation
- Describe the end products of Hb degradation
- Define hemoglobinopathies
- Enlist types of hemoglobinopathies
- Define Jaundice
- Describe Jaundice & its types

Definition:

- Red blood cells are continuously undergoing a hemolysis (breaking apart) process
- As the red blood cells disintegrate, the hemoglobin is degraded or broken into globin, the protein part, iron and heme.


RBC
effete rbc phagocytized by macrophages
Macrophage
degraded in phagolysosome
Heme + globin → aa
Porphyrin
Unconjugated bilirubin
Fe
added to proteins
stored (ferritin or hemosiderin)
Released to transferrin with aid of ceruloplasmin
Unconjugated bilirubin + albumin

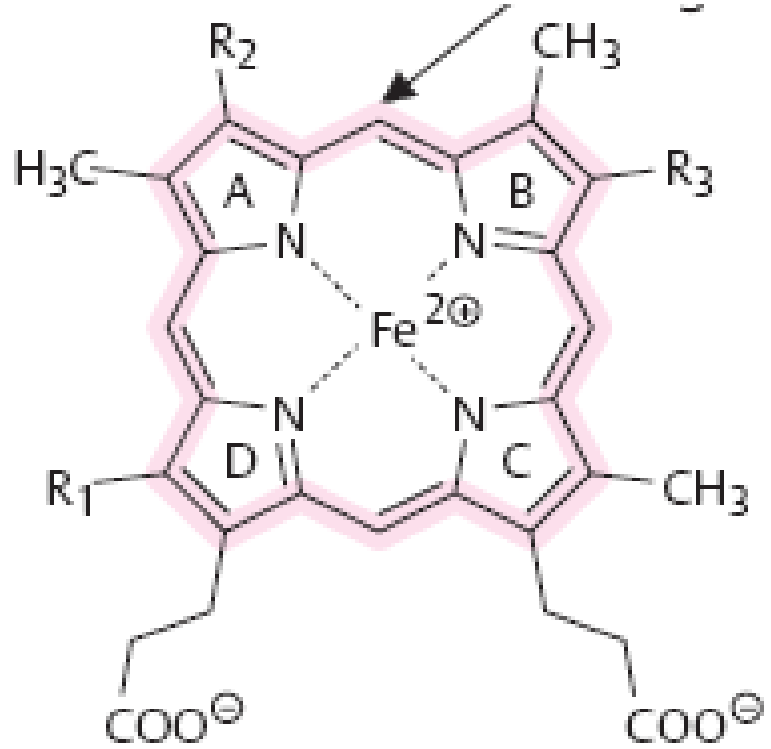

Breakdown of Hemoglobin
Hemoglobin
O_2, NADPH+H+
Heme oxygenase
NADP+, O_2
Fe^{+++}
Biliverdin
NADPH+H+
Bilirubin reductase
NADP+
Biliverdin
Amino acids
Reutilized

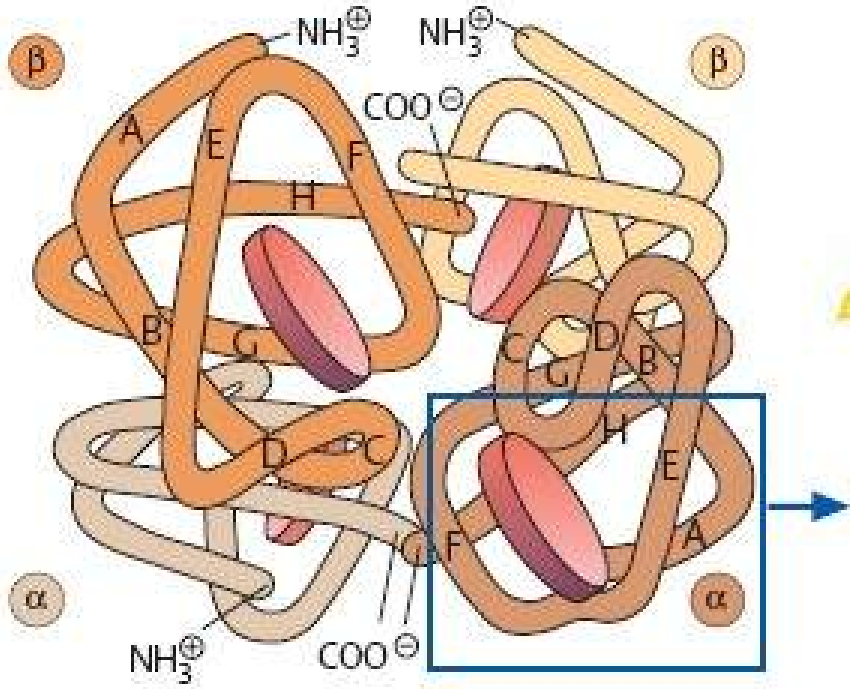
Sites

- Reticuloendothelial system
  - Liver
  - Spleen
  - Bone Marrow
  - 85% → RBCs, 15% → Immature RBCs

Fates

- Globin
  - Degraded
  - Amino acids reu
- Iron
  - Enters iron poo
  - Reused
- Porphyrin
  - Degraded





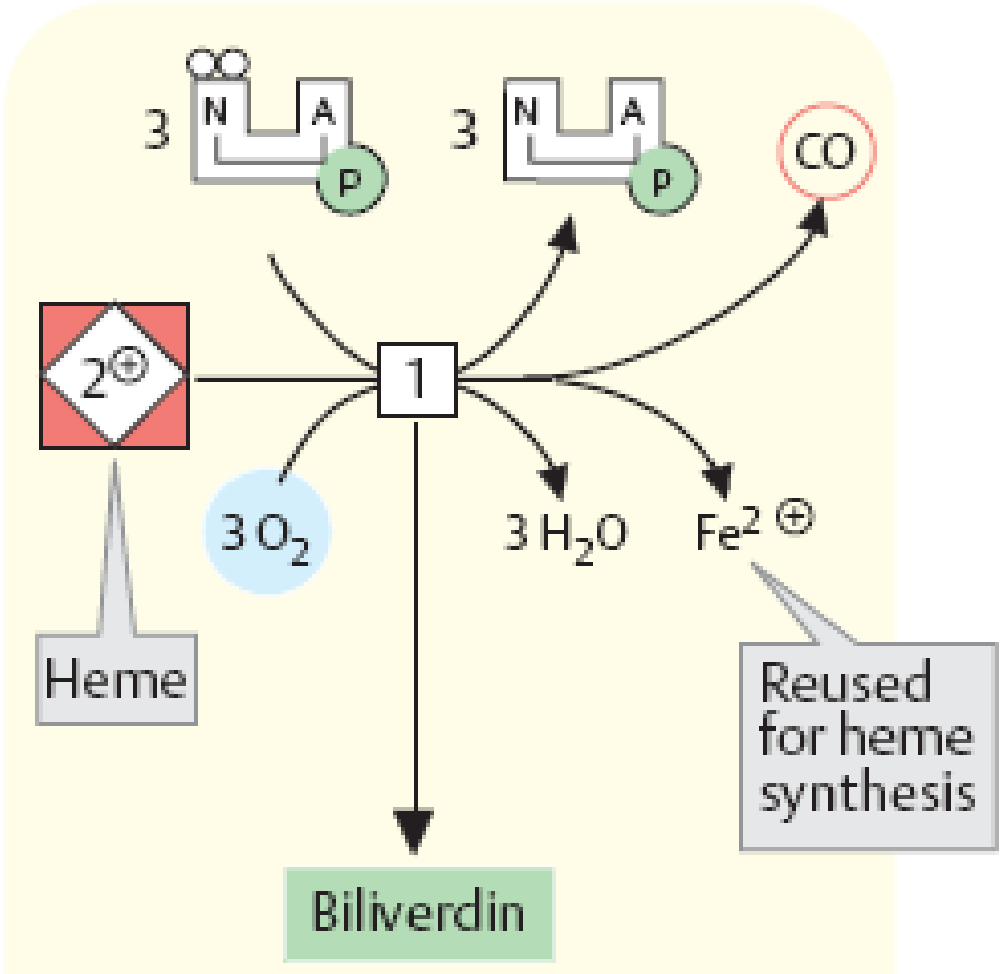
Hemoglobin A $(\alpha_2 \beta_2)$ M: 65 kDa

Steps

- Formation of bilirubin
- Uptake of bilirubin by the liver
- Formation of bilirubin diglucuronide
- Excretion of bilirubin into bile
- Formation of urobilins in the

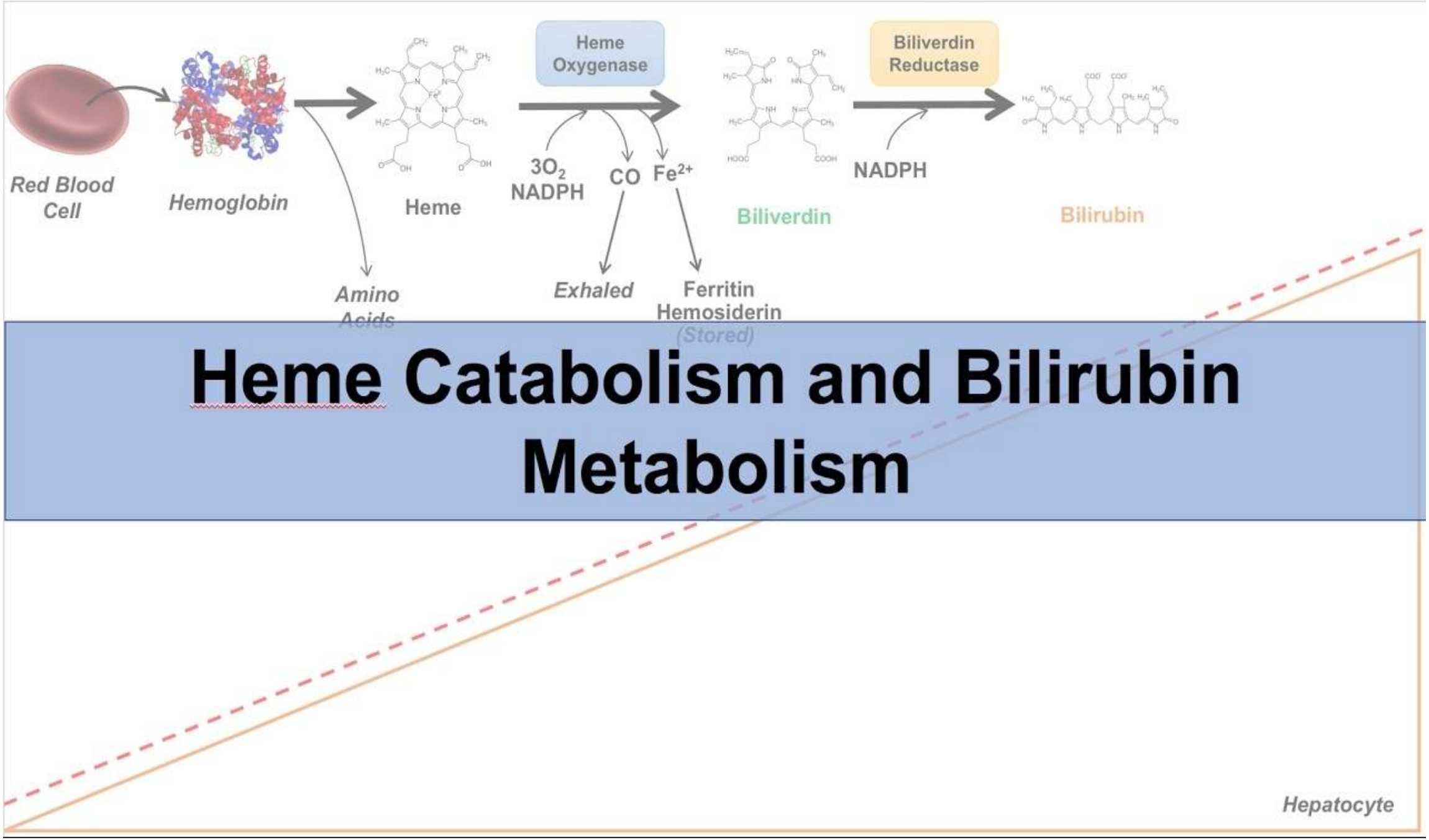
Requirements of Heme Degradation

- Enzyme
  - Heme Oxygenase System
- Molecular oxygen
- Reducing agent
  - Vitamin B3 (Niacin) derivative NADPH + H^+



Formation of bilirubin

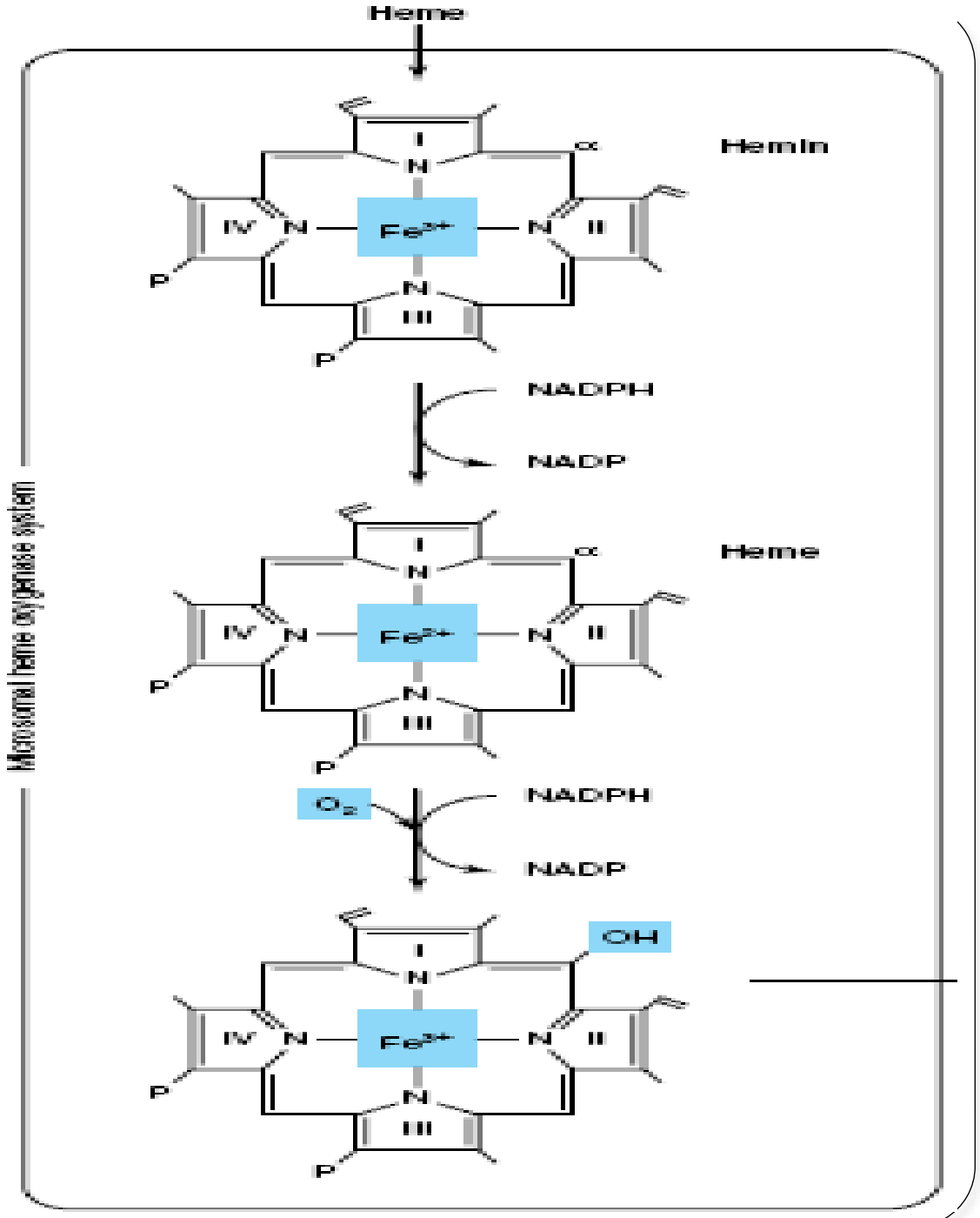
- Microsomal heme oxygenase system → RE → NADPH and O_2 → enzyme adds OH → bridge 2 pyrrol rings → Oxidation of Fe → cleavage of porphyrin ring → CO, Ferric ions
- Biliverdin (green pigment)
- Biliverdin is reduced → bilirubin (red orange)



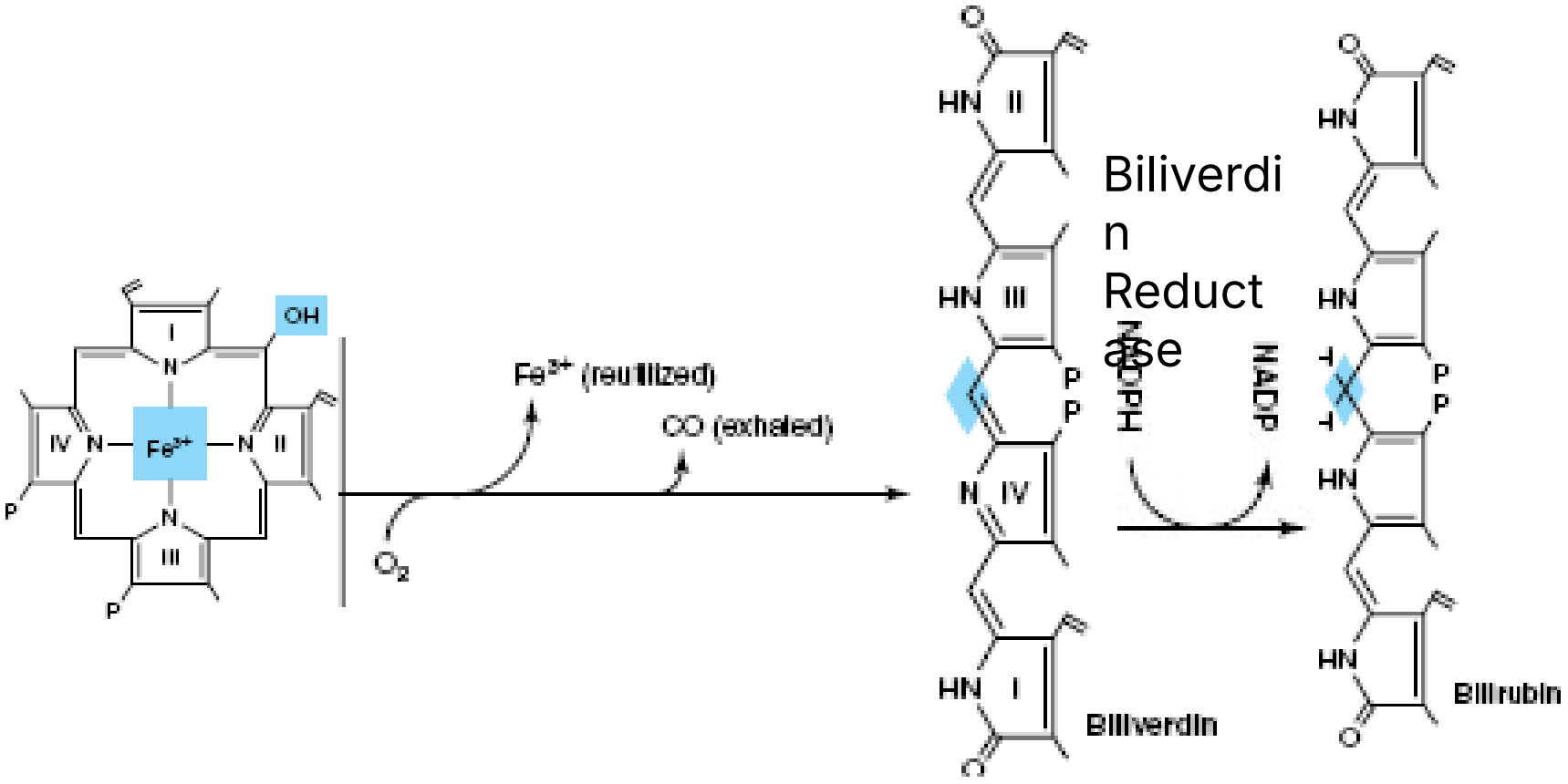
Heme Catabolism and Bilirubin Metabolism

Heme Oxygenase System

- Microsomes
- More than one enzymatic activity
- Hemin enters the system
- Substrate-inducible
- Hemin is reduced to Heme using $NADPH+H^{+}$
- Molecular oxygen is added
- Pyrrol I and II
- Fe^{+2} is oxidezed to Fe^{+3}
- Again oxygen is added
- Iron released
- CO is formed
- Tetrapyrrol opens
- Biliverdin released



Heme Degradation



Uptake of bilirubin by the liver

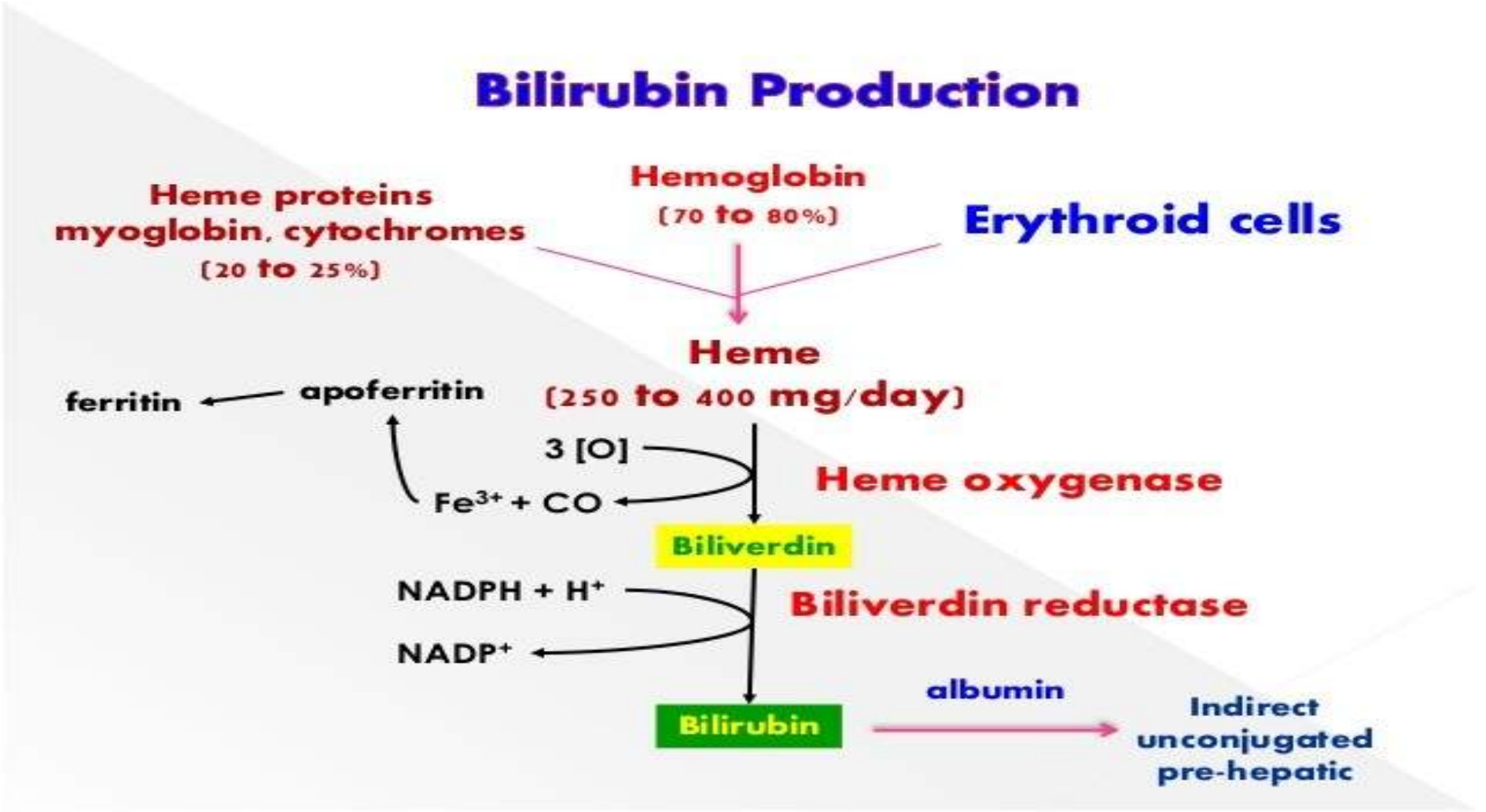
- Bilirubin + albumin → liver
- Bilirubin dissociates from albumin → hepatocytes + intracellular protein (Ligandin)

Formation of bilirubin diglucuronide

- Solubility of bilirubin increased by the addition of 2 molecules glucoronic acid
- Catalyzed → bilirubin glu-curonyl-transferase

Excreation of bilirubin into bile

- Bilirubin diglucuronide → Transported bile c



Formation of urobilins in the intestine

- Bilirubin glucuronide → hydrolyzed and reduced by gut bacteria → urobilinogen (colorless compound)
- Most urobilinogen is oxidized → intestinal bacteria → stercobilinogen (brown color)

Hemoglobinopathies

Definition

- Clinical diseases that result from a genetically determined abnormality of the **STRUCTURE** or **SYNTHESIS** of the hemoglobin molecule.
- The abnormality is associated with the globin chains
- The heme portion of the molecule is normal.

discussed in this article.

Hb name	Subunit composition	Mutation &/or clinical implication
Hb A	$Alpha_2beta_2$	n/a
Hb F	$Alpha_2gamma_2$	n/a
Hb A2	$Alpha_2delta_2$	Elevated in beta thalassemia
Hb Bart's	$Gamma_4$	Detected in pediatric patients with alpha thalassemia
Hb H	$Beta_4$	Detected in patients with alpha thalassemia
Hb S	$Alpha_2beta_2$	Beta mutation Glu6Val
Hb C	$Alpha_2beta_2$	Beta mutation Glu6Lys
Hb E	$Alpha_2beta_2$	Beta mutation Glu26Lys

Genotype	α-globin gene number[a]	Name	Phenotype
αα / αα	4	Normal state	None
αα / α–	3	Silent carrier	None (values for Hb and MCV may be near the lower limits of normal)
– – / αα or α– / α–	2	Thalassemia trait	Thalassemia minor: asymptomatic, mild microcytic anemia
– – / α–	1	Hb H disease	Thalassemia intermedia: mild to moderate microcytic anemia
– – / $\alpha^{CS}\alpha$	1	Hb H-Constant Spring	Thalassemia intermedia: moderate to severe microcytic anemia
– – / – –	0	Alpha thalassemia major	Thalassemia major: hydrops fetalis

[a]Number of normal alpha globin genes

Genotype	Hb A	Hb A2	Hb F
Normal	Normal	Normal	Normal
Silent Carrier	Normal	Normal	Normal
β Thalassemia Minor	Dec	Normal to Increased	Normal to Increased
β Thalassemia Intermedia	Dec	Normal to Increased	Increased
β Thalassemia Major	Dec	Usually Increased	Increased

Jaundice

Liver

- Major Role
- Reticuloendothelial system → synthesis and breakdown RBCs
- Liver cells metabolize, detoxify and excrete both endogenous and exogenous compounds.

Jaundice

- It is the yellow coloration of conjunctiva, mucous membrane and skin due to increased bilirubin levels.
- Its not a disease but a symptom of underlying disorder.

Classification

- 1929 → Rolieston and McNee
- 1964 → Maclagan
- 3 major groups

Jaundice

- Bilirubin → 1mg/dl (17.1 umol/L) → hyperbilirubinemia.
- D/t: ↑ production, excretion disturbed, obstruction to biliary duct.
- 2 – 2.5 mg/dl → diffuse into the tissue → yellow colour of skin.

Types

- Pre hepatic
- Hepatic
- Post hepatic

Causes

- Prehepatic: heamolysis
- Liver → conjugate and excrete >3000 mg bilirubin daily.
- Normal production of bilirubin 300mg/day
- Excessive degradation of RBCs
- ↑urobilinogen in urine, ↑unconjugated bilirubin in blood

Causes.....cont,

- Hepatic:
- Damage of liver cells.
- ↑unconjugated bilirubin, ↑urobilinogen
- Dark colour urine, pale, clay stool.
- ↑ ALT,AST levels
- Nausea, vomiting, anorexia.

Causes……cont,

- Post hepatic:
- Obstructive jaundice → obstruction in bile flow → stone, tumor
- Gastrointestinal pain, nausea
- ↑ Conjugated bilirubin, ALT ALK PO_4 levels

Types

- Unconjugated Type
- Hemolytic Anemias
- Neonatal Physiological jaundice
- Crigler – Najjar Syndrome
- Gilbert Syndrome
- Conjugated Type
- Dubin-Johnson Syndrome
- Rotor Syndrome

LFTs

- Measurements of blood components → provides a lead to the existence, extent type of liver damage.
- Bilirubin
- ALT
- Alk. PO4
- Total Protein

Investigations

- Obstruction to the biliary tract
- Acute hepatocellular damage
- Chronic liver disease
- Biochemical Markers:
- AST and ALT → Hepatocyte damage
- Bilirubin and Alkaline Phosphatase → presence of cholestasis
- Protein → liver synthetic capacity
- Disease progression → LFTs

Bilirubin

- Derived →Haem
- Adult → 450 umol daily
- Insoluble in water, bound to albumin
- Monoglucuronides → 25%
- Diglucuronides → 75%
- Functional constituent → bile salt
- Biliary tract → blocked → ↑Bilirubin in blood

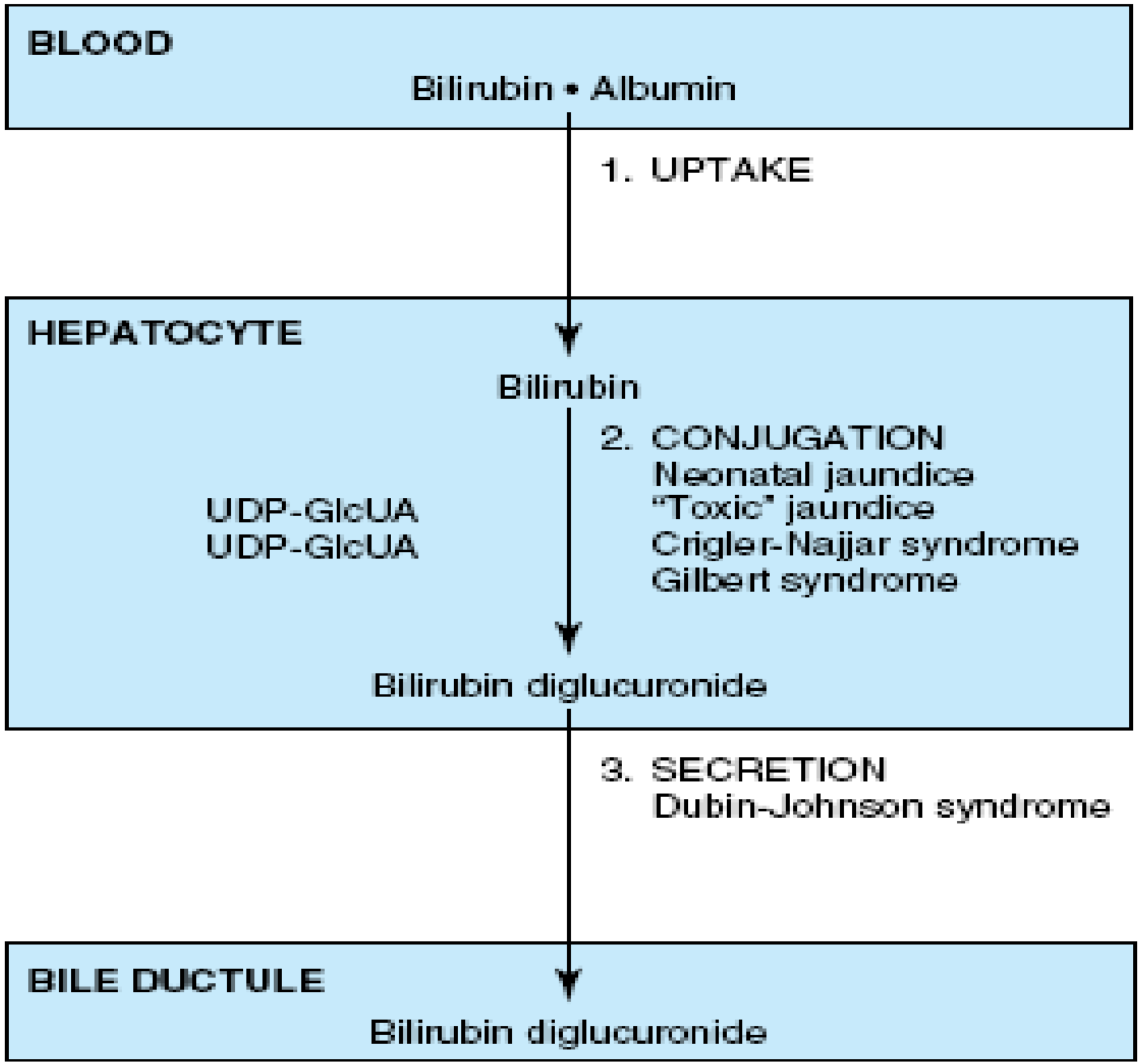

BLOOD
Bilirubin • Albumin
1. UPTAKE
HEPATOCYTE
Bilirubin
2. CONJUGATION
Neonatal jaundice
"Toxic" jaundice
Crigler-Najjar syndrome
Gilbert syndrome
UDP-GlcUA
UDP-GlcUA
Bilirubin diglucuronide
3. SECRETION
Dubin-Johnson syndrome
BILE DUCTULE
Bilirubin diglucuronide

Table 32–3. Laboratory results in normal patients and patients with three different causes of jaundice.

Condition	Serum Bilirubin	Urine Urobilinogen	Urine Bilirubin	Fecal Urobilinogen
Normal	Direct: 0.1–0.4 mg/dL Indirect: 0.2–0.7 mg/dL	0–4 mg/24 h	Absent	40–280 mg/24 h
Hemolytic anemia	↑ Indirect	Increased	Absent	Increased
Hepatitis	↑ Direct and Indirect	Decreased if micro-obstruction is present	Present if micro-obstruction occurs	Decreased
Obstructive jaundice[1]	↑ Direct	Absent	Present	Trace to absent

Aminotransferases (AST, ALT)

- Acute damage to hepatocytes → hepatitis, drug over dose , shock, sever hypoxia and acute cardiac failure

Alkaline Phosphatase

- Cholestasis
- Infiltrative disease of liver
- Cirrhosis
- γGT levels → microsomal enzyme

Plasma Proteins

- Albumin → long biological ½ life in plasma (20 days).
- Hypoalbuminaemia → Advanced chronic liver disease, severe acute liver damage
- Globulin → severity of liver disease

References

- Harper 27th edi 31 287
- Lippencott’s 21 280


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Thank
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